

Notes: Jones (AEJ: Macroeconomics, 2011)

Intermediate Goods and Weak Links in the Theory of Economic Development

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1 Big picture

- **Question.** How can relatively modest micro-level distortions generate enormous cross-country income differences?
- **Core idea.** Development accounting should not treat output as if it is produced only from primary factors. Modern economies are dense input-output networks. Distortions to one sector propagate through intermediate goods and become amplified.
- **Two old development ideas revived.** The paper returns to **linkages** and **complementarity**. Linkages work through intermediate goods, while complementarity means that many tasks or inputs are jointly necessary for high productivity.
- **Intermediate-goods multiplier.** If intermediate inputs are a large share of gross output, then a productivity or distortion shock affects production repeatedly along the supply chain. The multiplier is analogous to the capital multiplier in the Solow model, but quantitatively larger because the intermediate share is about one-half.
- **Weak-link logic.** If production requires many complementary inputs, then a failure in one sector can sharply reduce output elsewhere. Electricity, finance, transport, raw materials, legal permits, and security are all examples of weak links.
- **Main quantitative result.** Calibrating micro distortions and plant-level productivity dispersion using Hsieh and Klenow statistics, the model can amplify rich-poor income ratios by factors typically between 2 and 4, and as high as 6.2 in the baseline scenario.
- **Development accounting implication.** Aggregate TFP differences may reflect amplified micro distortions. A given wedge at the plant or sector level can generate much larger aggregate income effects once intermediate goods are included.
- **Policy implication.** The model explains why development may be hard: fixing only one bottleneck may not deliver large gains if other complementary inputs remain distorted. At the same time, broad reductions in distortions can have large aggregate effects.
- **Takeaway.** The paper provides a production-network mechanism through which misallocation, theft, taxes, corruption, markups, and missing institutions can have much larger effects on income than standard neoclassical models suggest.

2 Incremental contribution of the paper

2.1 Relative to standard development accounting

Standard development accounting typically decomposes income differences into physical capital, human capital, and residual TFP. With a capital share near one-third, the standard capital multiplier is too small to explain very large income gaps unless residual TFP differences are huge. Jones adds intermediate goods as produced inputs and shows that they raise the effective produced-input share.

$$\beta = \alpha(1 - \sigma) + \sigma, \tag{1}$$

where α is the capital exponent and σ is the intermediate-goods exponent. If $\alpha = 1/3$ and $\sigma = 1/2$, then $\beta = 2/3$, and the multiplier becomes approximately

$$\frac{1}{1 - \beta} = 3. \tag{2}$$

Interpretation: The incremental contribution is to show that intermediate inputs are not a second-order detail. They move the economy from a low-multiplier world to a high-multiplier world.

2.2 Relative to input-output economics

Input-output economics has long emphasized sectoral linkages and multipliers. Jones imports these ideas into modern macroeconomic development theory and integrates them with capital accumulation, human capital, plant-level wedges, and development accounting.

Interpretation: The paper bridges an older Leontief-Hirschman input-output tradition and the modern misallocation/development-accounting literature.

2.3 Relative to weak-link theories

Michael Kremer’s O-ring theory highlights the idea that output can be limited by the weakest task in a production chain. Jones embeds weak links in a general equilibrium production framework with substitution and intermediate goods, showing when weak-link effects are large and when substitution mitigates them.

Interpretation: The incremental contribution is not merely to say that weak links matter, but to quantify their role when firms can substitute capital and labor away from bad sectors.

2.4 Relative to Hsieh and Klenow style misallocation

Hsieh and Klenow measure plant-level productivity and wedges in China, India, and the United States. Jones uses their micro facts as inputs to a production-network model and asks how much larger the macro effects become once intermediate goods and complementarities are added.

Interpretation: The paper changes the interpretation of micro wedges: the same measured wedges produce much larger income effects in an economy with intermediate-good linkages.

3 Model intuition: linkages through intermediate goods

3.1 Simple production example

Start with gross output Q_t produced using capital K_t , labor L_t , and intermediate goods X_t :

$$Q_t = A(K_t^\alpha L_t^{1-\alpha})^{1-\sigma} X_t^\sigma, \quad (3)$$

where σ is the intermediate-goods share of gross output.

Output can be consumed, invested, or used as intermediate input. If a constant fraction χ of gross output is used as intermediate input, then GDP is

$$Y_t = (1 - \chi)Q_t. \quad (4)$$

Capital evolves as

$$K_{t+1} = I_t + (1 - \delta)K_t, \quad (5)$$

and intermediate goods evolve as

$$X_{t+1} = \chi Q_t. \quad (6)$$

3.2 Multiplier without and with intermediate goods

In the steady state, the effect of productivity and misallocation on output is amplified by both capital accumulation and intermediate-good feedback. Without intermediates, the multiplier is the familiar

$$\frac{1}{1 - \alpha}. \quad (7)$$

With intermediate goods, the multiplier becomes

$$\frac{1}{(1 - \alpha)(1 - \sigma)}. \quad (8)$$

Equivalently, defining the total produced-input share as

$$\beta = \alpha(1 - \sigma) + \sigma, \quad (9)$$

the multiplier is

$$\frac{1}{1 - \beta}. \quad (10)$$

Interpretation: The multiplier becomes large because intermediate goods are produced factors that are used up within the same production process. A distortion can be paid repeatedly as goods pass through the input-output chain.

3.3 Quantitative intuition

With $\alpha = 1/3$ and $\sigma = 1/2$:

$$\beta = \frac{1}{3} \left(1 - \frac{1}{2} \right) + \frac{1}{2} = \frac{2}{3}. \quad (11)$$

Thus the produced-input multiplier is 3. A doubling of fundamental productivity raises output by a factor of $2^3 = 8$ in the intermediate-goods economy, compared with $2^{3/2} \approx 2.8$ in a model with only capital accumulation.

Interpretation: This is the paper's main mechanical force. Intermediate goods turn moderate productivity or wedge differences into large income differences.

4 Model environment

4.1 Goods and sectoral technologies

There is a continuum of goods indexed by $i \in [0, 1]$. Each sector produces

$$q_i = A_i (K_i^\alpha H_i^{1-\alpha})^{1-\sigma} X_i^\sigma, \quad (12)$$

where A_i is sectoral productivity, K_i is physical capital, H_i is human capital, and X_i is the aggregate intermediate input used by sector i .

Each good can be used as final consumption c_i or as an intermediate input z_i :

$$c_i + z_i = q_i. \quad (13)$$

4.2 Final goods aggregator

Final goods aggregate into GDP as

$$Y = \left(\int_0^1 c_i^\theta di \right)^{1/\theta}, \quad 0 < \theta < 1. \quad (14)$$

This implies an elasticity of substitution across final goods greater than one.

Interpretation: Final consumption goods are substitutable. If one consumption variety is bad, consumers can substitute toward others.

4.3 Intermediate-goods aggregator

Intermediate inputs aggregate as

$$X = \left(\int_0^1 z_i^\rho di \right)^{1/\rho}, \quad \rho < 0. \quad (15)$$

This implies an elasticity of substitution below one. Intermediate goods are complementary.

Interpretation: This is where weak links enter. If a key intermediate input is low productivity or distorted, it reduces the productivity of many other sectors.

4.4 Resource constraints

Physical and human capital constraints are

$$\int_0^1 K_i di \leq K, \quad (16)$$

$$\int_0^1 H_i di \leq H = hL, \quad (17)$$

with labor normalized to $L = 1$. Capital evolves as

$$\dot{K} = I - \delta K, \quad (18)$$

and the final good resource constraint is

$$C + I \leq Y. \quad (19)$$

5 Symmetric allocation and benchmark results

5.1 Symmetric allocation

A useful benchmark assumes that all sectors receive the same quantities of capital, human capital, and intermediate goods, and that fixed fractions of output are invested or used as intermediates. Under this allocation, GDP is

$$Y = \zeta(z) (S_\rho^\sigma S_\theta^{1-\sigma}) K^\alpha H^{1-\alpha}, \quad (20)$$

where

$$S_\rho = \left(\int_0^1 A_i^{\rho/(1-\sigma)} di \right)^{(1-\sigma)/\rho}, \quad (21)$$

and S_θ is defined analogously.

Interpretation: The same productivity distribution enters differently depending on whether goods are used as intermediate inputs or final goods. Complementarity makes low-productivity sectors more important.

5.2 Weak links versus superstar effects

Because $\rho < 0$, the intermediate-goods productivity aggregator behaves like a weak-link aggregator. Low productivity in one intermediate sector can lower aggregate productivity. Because $0 < \theta < 1$, the final-goods aggregator emphasizes superstar sectors more than weak links.

Interpretation: The same distribution of sectoral productivity has two faces: weak links in production and superstar goods in consumption.

6 Competitive equilibrium with wedges

6.1 Wedges

The competitive equilibrium introduces sector-specific wedges τ_i that distort production. These wedges can represent theft, taxes, bribes, markups, corruption, misallocation, or any policy or institutional friction that reduces effective output.

A simple way to view the wedge is that only a fraction $1 - \tau_i$ of gross output is effectively available to the firm or economy.

Interpretation: The paper takes these wedges as given and asks how much they are amplified by intermediate-goods linkages.

6.2 Steady-state multiplier

The steady state of the competitive equilibrium shows that GDP per worker depends on wedge-adjusted aggregate productivity and the fundamental multiplier. The exact expression is complicated, but the key structure is

$$y^* = \text{constant} \times [\text{wedge-adjusted aggregate productivity}]^{1/(1-\beta)} h. \quad (22)$$

Interpretation: Wedges do not simply reduce the sector where they occur. They feed through intermediate goods and capital accumulation to lower the entire economy's level of income.

6.3 Symmetric wedges

If the same wedge applies to every sector, $\tau_i = \bar{\tau}$, then the distortion still matters because intermediate goods and capital are produced inputs. A good used for final consumption is distorted once, but an intermediate good is distorted again each time it is used in further production.

Interpretation: This is a dynamic Diamond-Mirrlees logic. Taxing intermediate goods or produced inputs is especially costly because the distortion is compounded along the production chain.

6.4 Random wedges

When productivity and wedges are jointly lognormal, the paper derives a tractable expression for log income. In the Cobb-Douglas limit, the average wedge and the variance of wedges are both amplified by the model's fundamental multiplier:

$$y^* \propto (1 - \bar{\tau})^{\beta/(1-\beta)} \exp \left[-\frac{\eta}{2(1-\beta)} \text{Var}(\log(1 - \tau_i)) \right]. \quad (23)$$

Interpretation: Variation in distortions across plants or sectors creates misallocation. The production network amplifies both average distortions and the dispersion of distortions.

7 Development accounting

7.1 Intermediate-goods share

Jones uses OECD input-output data to measure the intermediate-goods share of gross output. This is the crucial parameter σ .

Read:

- Figure 1 reports the intermediate goods share of gross output for many countries.
- The United States and India have shares around 47 percent.
- Japan's share is about 52 percent.
- China has the highest share in the figure, about 68 percent.
- Across the 35 countries, the average intermediate-goods share is about 52.6 percent, with a standard deviation around 6 percent.

Economic message: The data justify the benchmark $\sigma = 1/2$. Intermediate goods are roughly half of gross output, making the input-output multiplier quantitatively first order.

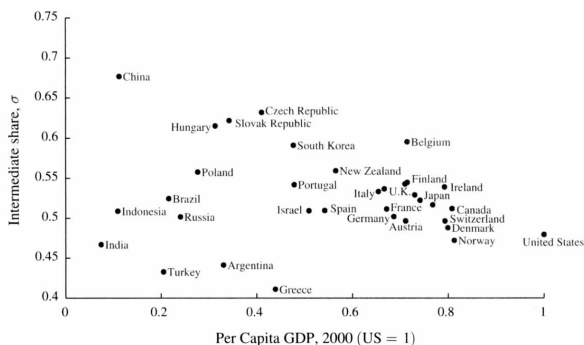


FIGURE 1. THE INTERMEDIATE GOODS SHARE OF GROSS OUTPUT

7.2 Numerical exercise using Hsieh-Klenow statistics

The paper uses Hsieh and Klenow plant-level estimates for China, India, and the United States. The exercise compares a “rich” economy calibrated to the United States and a “poor” economy calibrated to China/India.

The goal is to see how much rich-poor income ratios change when intermediate-goods linkages and complementarity are added.

8 Quantitative results

8.1 Table 1: output per worker ratios

Read:

- Table 1 compares income ratios with no intermediate goods ($\sigma = 0$) and with the baseline intermediate share ($\sigma = 1/2$).
- In the baseline scenario, the no-intermediate model generates a rich-poor income ratio of 4.7.
- With intermediate goods and weak links, the ratio rises to 29.0.
- The amplification factor is therefore 6.2.
- Across scenarios, amplification factors range from 1.3 to 6.2.

Economic message: For the same micro distortions, adding intermediate goods and weak links can multiply the implied income gap several times over.

8.2 Table 2: robustness to complementarity

Read:

- Table 2 varies the degree of complementarity among intermediate goods.
- It compares Cobb-Douglas ($\rho = 0$), the baseline ($\rho = -1$), and near-Leontief ($\rho = -100$).
- Amplification factors remain large across cases.
- The difference between Cobb-Douglas and near-Leontief is not as large as a simple weak-link intuition might suggest.

Economic message: Complementarity matters, but substitution through capital and labor prevents the weak-link effect from becoming arbitrarily large.

TABLE 1—OUTPUT PER WORKER RATIOS USING THE HSIEH-KLENOW STATISTICS

Scenario	Description	Average TFP in poor country $\sigma = 0$	No interme- diate goods $\sigma = 0$	Baseline case $\sigma = 1/2$	Amplification factor
1.	Baseline	0.604	4.7	29.0	6.2
2.	Identical TFPs	1.000	3.4	4.3	1.3
3.	$\nu_a^{rich} = \nu_a^{poor} = 0.84$	0.800	4.8	8.4	1.8
4.	$\nu_a^{rich} = \nu_a^{poor} = 0.5$	0.800	4.1	7.7	1.9
5.	$\nu_a^{rich} = 0.5, \nu_a^{poor} = 0.75$	0.654	4.9	16.9	3.4
6.	Same as 5, but $\nu_{aw} = 0$	0.654	3.5	14.2	4.0
7.	Same as 6, but $\bar{\tau}^{poor} = 0$	0.654	3.1	10.3	3.3

Notes: The two main columns of the table report income ratios between a “rich” and a “poor” (e.g., China or India) country based on Proposition 5. For comparison, the income ratios between the United States and China/India were 8.9 and 14.6 in 2000. The “amplification factor” column shows the ratio of the two previous columns—that is, the overall amplification from having $\sigma = 1/2$. Baseline parameter values are $\theta = 2/3, \rho = -1, \bar{K}^{rich}/\bar{K}^{poor} = 1.52, \nu_a^{rich} = 0.84$ (US), $\nu_a^{poor} = 1.23$ (India), $\nu_a^{rich} = 0.45, \nu_a^{poor} = 0.68, \mu_a^{rich} = -0.5 \times (\nu_a^{rich})^2, \mu_a^{poor} = -0.5(\nu_a^{poor})^2, \mu_{aw}^{poor} = \log(1 - \bar{\tau}^{poor}) = 0.5(\nu_a^{poor})^2, \nu_{aw}^{rich} = 0$ and $\nu_{aw}^{poor} = -0.647 \cdot \nu_a^{poor} \mu_{aw}^{poor}$. Most of these numbers are taken from Hsieh and Klenow (2009), tables 1–2. The main exception is the value for μ_{aw} , on which Hsieh and Klenow are silent; we assume the average distortion rate ($\bar{\tau}^{poor}$) is $1/5$ in Scenarios 1–6 and is 0 in Scenario 7. The correlation coefficient 0.647 is based on Table 4 of the NBER working paper version, Hsieh and Klenow (2007), not reported in the published version. In Scenarios 1, 5, 6, and 7, μ_a^{poor} is chosen so that TFP at the ninety-ninth percentile is the same in the rich and poor countries (otherwise, the higher variance in the poor country would lead firms at the top of the distribution to be substantially more productive than those in the rich country). Scenarios 3 and 4 set the simple average of TFP across firms in the poor country at 0.8.

Table 1: output per worker ratios

TABLE 2—ROBUSTNESS RESULTS WITH THE HSIEH-KLENOW STATISTICS

Scenario	Description	Amplification factors		
		Cobb-Douglas $\rho = 0$	Baseline $\rho = -1$	“Leontief” $\rho = -100$
1.	Baseline	5.5	6.2	6.8
2.	Identical TFPs	1.4	1.3	1.1
3.	$\nu_a^{rich} = \nu_a^{poor} = 0.84$	2.0	1.8	1.5
4.	$\nu_a^{rich} = \nu_a^{poor} = 0.5$	2.0	1.9	1.8
5.	$\nu_a^{rich} = 0.5, \nu_a^{poor} = 0.75$	3.4	3.4	3.5
6.	Same as 5, but $\nu_{aw} = 0$	3.4	4.0	4.9
7.	Same as 6, but $\bar{\tau}^{poor} = 0$	2.9	3.3	3.8

Notes: The main columns report the “amplification factor,” i.e. the proportional increase in the rich/poor income ratio that occurs when going from $\sigma = 0$ to $\sigma = 1/2$. There are no robustness results associated with varying θ because the amplification factors are invariant to this parameter. See notes to Table 1.

Table 2: robustness to complementarity

9 Interpretation and mechanisms

9.1 Theft example

Jones uses theft as an intuition. If half of output is stolen from each firm, the direct effect is large. But in an intermediate-goods economy, the stolen output may be stolen repeatedly through the supply chain. Steel is stolen from the steel plant, then a distorted steel input raises the cost of car production, then the car service raises the cost of other goods.

Interpretation: A wedge on gross output is paid multiple times in a production network. This is the input-output multiplier in intuitive form.

9.2 Why there is no single magic bullet

A high multiplier could appear to imply that small reforms have huge effects. Jones argues that complementarity mitigates this. If many weak links exist, fixing only one may not greatly raise output because other links remain weak.

Interpretation: The model explains both large income gaps and the difficulty of development policy. Broad improvement across many margins is needed.

9.3 Connection to institutions

The paper does not model why poor countries have worse wedges. Instead, it complements institutional theories. Political economy explains why distortions arise; Jones explains how given distortions become amplified into large income differences.

Interpretation: The contribution is mechanical but essential. Institutions matter more when production networks amplify their distortions.

10 Identification and model discipline

10.1 What is disciplined by data

- The intermediate-goods share σ is disciplined by input-output tables.
- Plant-level dispersion in productivity and wedges is disciplined by Hsieh and Klenow statistics.

- Human capital differences are summarized using educational attainment and returns to schooling.
- Elasticities of substitution are varied to test robustness.

10.2 What is not identified

- The model does not identify the political or institutional origin of wedges.
- It does not estimate country-specific input-output networks in full detail.
- It abstracts from endogenous human capital accumulation, innovation, firm entry, and political economy.
- The Hsieh-Klenow productivity/wedge measures may contain measurement error.

Interpretation: The paper is best read as a theory-and-accounting exercise: if micro distortions of the measured magnitude exist, intermediate goods and weak links can make their aggregate consequences much larger.

11 Short summary

- The paper studies how intermediate-goods linkages and weak links amplify distortions in development accounting.
- Intermediate goods are about half of gross output, so they create a large multiplier.
- Weak links arise because intermediate inputs are complementary in production.
- Sectoral productivity and wedge dispersion matter more when goods are used as intermediate inputs.
- Symmetric wedges are amplified because intermediate goods and capital are produced inputs.
- Random wedges generate misallocation, and the variance of wedges is amplified by the production network.
- Using Hsieh-Klenow plant-level statistics, the model can multiply rich-poor income ratios by factors between roughly 2 and 4 in many scenarios and 6.2 in the baseline.
- Complementarity matters, but substitution through capital and labor limits the extreme weak-link effect.
- The paper complements institutional theories by explaining how given distortions generate large income differences.

12 Conclusion

12.1 Core conclusion

Jones shows that input-output linkages and weak-link complementarities substantially amplify the macroeconomic impact of micro distortions. Because intermediate goods are roughly half of gross output, distortions can be compounded through production chains, making rich-poor income differences much larger than in a standard neoclassical model.

12.2 Economic intuition

In a standard model, a productivity loss in one sector affects that sector and then capital accumulation. In a production-network model, the loss also raises the cost of intermediate inputs for other sectors, which then lowers their output and feeds back again. This recursive structure creates a powerful multiplier. Complementarity means that badly performing sectors can become weak links, further lowering aggregate output.

12.3 Takeaway

The paper's central lesson is that development is a network problem. Poor institutions, corruption, theft, taxes, markups, or misallocation matter not only where they occur but everywhere they are transmitted through input-output chains. Understanding large income differences therefore requires both an explanation of distortions and a multiplier mechanism that turns them into large aggregate losses.